

Fagnano's Problem

The mathematical sciences particularly exhibit order, symmetry, and limitation; and these are the greatest forms of the beautiful. – Aristotle

CHAPTER 16

The More Things Change...

A **geometric transformation** is a rule we apply to a geometric figure that usually results in the figure being moved or changed in some way. In this chapter we will explore a few basic types of transformations by sliding, spinning, and flipping figures. As we'll see, these transformations change the locations of figures to which they are applied, as opposed to turning them into entirely different figures.

16.1 Translations

When we apply a **translation** to a figure, we simply slide it a specified distance in a given direction. For example A' and B' below are the result of translating A and B , respectively, in the direction and by the distance suggested by the arrows.

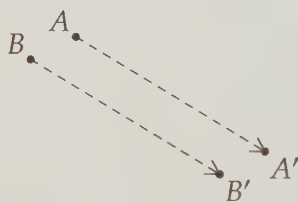


Figure 16.1: A Translation

We use some special terms when performing transformations. We say that A' is the **image** of A under the translation shown above. We can also say that the translation **maps** A to A' .

Of course, we can **translate** more than just points. We can translate any figure – a segment, a line,

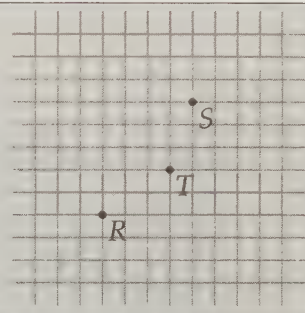
a triangle, etc. – and the result will be congruent to our original figure. This is one of those ‘obvious’ things that we won’t worry about proving right now. We’ll look at proving ‘obvious’ facts like these in the next book in this series, *Intermediate Geometry*.

Problems

Problem 16.1: A **fixed point** of a transformation is a point that is its own image. The **identity** transformation is the transformation that maps every point to itself. In other words, the identity is the ‘do nothing’ transformation. Can a translation that is not the identity have a fixed point?

Problem 16.2:

- Locate point U on the grid shown such that $RSTU$ is a parallelogram.
- Is the U you found the only point that can be combined with R , S , and T to form the vertices of a parallelogram? If not, find the others.



Translations are the simplest transformations, so there aren’t many interesting problems involving translations. We’ll use these simple transformations to introduce a couple more terms.

Problem 16.1: A **fixed point** of a transformation is a point that is its own image. The **identity** transformation is the transformation that maps every point to itself. In other words, the identity is the ‘do nothing’ transformation. Can a translation that is not the identity have a fixed point?

Solution for Problem 16.1: In a translation, every point is moved the same distance in the same direction. If there is a fixed point, then that point is not moved at all. Since all points must move the same distance in the same direction, this means none of the points is moved by the translation. Hence, the only translation that has a fixed point is the identity. $\square$

Concept: Understanding transformations is often more about learning what stays the same rather than what changes. In the three basic transformations we’ll study in this chapter, everything will stay the same for each transformed figure except its location and orientation. That may sound like a lot to change, but it really isn’t much. Line segments remain line segments, circles remain circles, points that are one unit apart remain one unit apart, and so on.

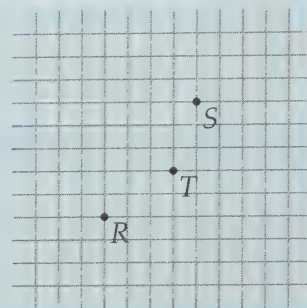
Here’s one example of using translations to solve a problem:

Extra! “Obvious” is the most dangerous word in mathematics.

–Eric Temple Bell

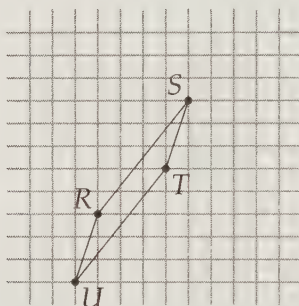
Problem 16.2:

- (a) Locate point U on the grid shown such that $RSTU$ is a parallelogram.
- (b) Is the U you found the only point that can be combined with R , S , and T to form the vertices of a parallelogram? If not, find the others.



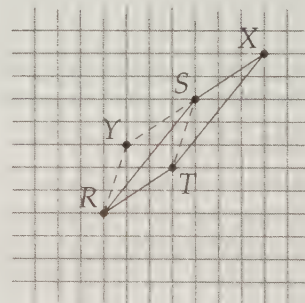
Solution for Problem 16.2:

- (a) If $RSTU$ is a parallelogram, then $\overline{SR} \parallel \overline{TU}$ and $SR = TU$. In other words, the same translation that maps S to R must map T to U . Since we must go down 5 and left 4 to get from S to R , we must go down 5 and left 4 to get from T to U . Therefore, U is at the point shown in the diagram below.



- (b) In the first part, we 'completed' the parallelogram that has $\angle RST$ as an angle. We could instead have chosen $\angle SRT$ or $\angle STR$ as an angle of the parallelogram. Let points X and Y , 'complete' parallelograms $SRTX$ and $STRY$, respectively.

To complete parallelogram $SRTX$, we note that we must go right 3 and up 2 to get from R to T . Therefore, we go right 3 and up 2 from S to get to X . Similarly, to complete $STRY$, we see that we go left 3 and down 2 to get from T to R , so we must go left 3 and down 2 from S to find Y . Our other two parallelograms are shown in the diagram.



□

Exercises

16.1.1 $ABCDEF$ is a regular hexagon. Why must the translation that maps E to C also map F to B ?
Hints: 26, 32

Extra! Somebody came up to me after a talk I had given, and said, "You make mathematics seem like fun." I was inspired to reply, "If it isn't fun, why do it?"

—Ralph Boas

16.2 Rotations

When we **rotate** a figure, we spin it by some angle about some point. In other words, the image of point A under a **rotation** of angle θ counterclockwise about point O is the point A' such that $OA = OA'$ and $\angle AOA' = \theta$, as shown at right. There are two points that satisfy this definition of A' , which is why we must specify the direction of the rotation as clockwise or counterclockwise. Point O is the center of the rotation, and θ is the angle of rotation.



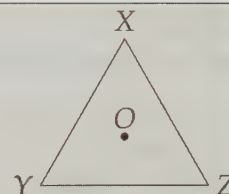
As with translations, the image of any figure upon rotation is congruent to the original figure. This is one of those 'obvious' facts that you might try proving on your own for specific shapes like line segments or circles.

Sidenote: The contributions of the ancient Greeks to mathematics have been immortalized in many ways. One of these is shown in our introduction of rotation. It's a common convention to use Greek letters as variables for angle measures. Above, we use the letter θ , or 'theta.' The letter ϕ is also often used for angles, and sometimes the Greek letters α , β , and γ are used to stand for the angles of a generic triangle $\triangle ABC$. Greek letters are also fun to write!

Problems

Problem 16.3: Let O be the center of equilateral $\triangle XYZ$.

- Through what angles between 0° and 360° can we rotate Y about Z such that the result is point X ?
- Through what angles between 0° and 360° can we rotate X about O such that the result is point Y ?
- Is there a rotation about Y that maps O to Z ?



Problem 16.4: Through how many different positive angles θ less than 360° is it possible to rotate a regular dodecagon clockwise about its center such that its image coincides with the original dodecagon?

Problem 16.5: After I draw a figure, I rotate it 48° about some point P . If the image of this rotation coincides exactly with the original figure, is it true that the image of a rotation of the figure 72° about P must also coincide with the figure?

We'll explore rotations with a few basic problems. As you'll see, rotations are all about angles.

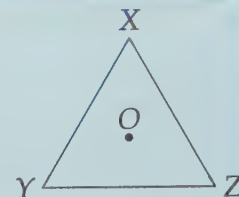
Extra! *The more things change, the more they are the same.*



—Alphonse Karr

Problem 16.3: Let O be the center of equilateral $\triangle XYZ$.

- Through what angle(s) between 0° and 360° can we rotate Y about Z such that the result is point X ?
- Through what angle(s) between 0° and 360° can we rotate X about O such that the result is point Y ?
- Is there a rotation about Y that maps O to Z ?



Solution for Problem 16.3:

- Since $YZ = XZ$ and $\angle YZX = 60^\circ$, a 60° rotation clockwise about Z takes Y to X . Of course, a 300° rotation the other way will also do the job!
- Since O is the center of equilateral $\triangle XYZ$, $\overline{XO}$ and $\overline{YO}$ bisect $\angle ZXY$ and $\angle ZYX$, respectively. Therefore, $\angle YXO = \angle OYX = 30^\circ$, so $\angle XOY = 180^\circ - \angle YXO - \angle OYX = 120^\circ$. Since $OX = OY$ and $\angle XOY = 120^\circ$, if we rotate X 120° counterclockwise about O , it is mapped to point Y . Note that we can also rotate X 240° clockwise around O to get to Y .
- By the definition of rotation, every rotation about Y maps O to a point that is as far from Y as O is. Since $\overline{YO}$ and $\overline{YZ}$ are not equal in length (make sure you see why!), it is impossible for a rotation about Y to map O to Z .

□

Problem 16.4: Through how many different positive angles θ less than 360° is it possible to rotate a regular dodecagon clockwise about its center such that its image coincides with the original dodecagon?

Solution for Problem 16.4: Let $ABCDEFGHIJKL$ be our regular dodecagon and O its center. Each of the interior angles has measure $180^\circ - 360^\circ/12 = 150^\circ$, and each segment from a vertex to the center bisects an interior angle, so $\angle OAB = \angle OBA = 75^\circ$. Therefore, $\angle AOB = 30^\circ$. So, if we rotate the dodecagon 30° , every vertex will be 'shifted over' by one vertex, and the dodecagon will map to itself.



Because a rotation of 30° works, we know a rotation of $30^\circ + 30^\circ = 2(30^\circ) = 60^\circ$ will work too. A 60° rotation is just the result of two 30° rotations, and each of these maps the dodecagon to itself. We could also think of this as shifting each vertex over two vertices. Similarly, a rotation of each multiple of 30° will map the dodecagon to itself. This gives us 11 positive angle rotations less than 360° about the center that will take the dodecagon to itself.

It is impossible for two different clockwise rotations between 0° and 360° about O to map A to the same vertex, and we can't map A to itself with such a rotation. Therefore, we know that the 11 rotations we've found are the only clockwise rotations between 0° and 360° that map the dodecagon to itself. □

Problem 16.5: After I draw a figure, I rotate it 48° about some point P . If the image of this rotation coincides exactly with the original figure, is it true that the image of a rotation of the figure 72° about P must also coincide with the figure?

Solution for Problem 16.5: We know that rotating the figure 48° leaves the figure unchanged, so to think about what happens when we rotate by 72° , we start by finding what other rotations leave the figure unchanged. Spinning the figure by 48° twice will leave it unchanged, so a 96° rotation leaves the figure unchanged. Similarly, other angles of rotation about P that don't change the figure are $3(48^\circ) = 144^\circ, 4(48^\circ) = 192^\circ, 240^\circ, 288^\circ, 336^\circ, 384^\circ, 432^\circ, 480^\circ, \dots$ Taking a look at those 'over 360° ' rotations, we see that one of them is 432° . A 432° rotation is equivalent to a $432^\circ - 360^\circ = 72^\circ$ rotation. Therefore, a 72° rotation will also coincide with the original figure. $\square$

Exercises

16.2.1 $ABCDEFGH$ is a regular octagon, with the vertices labeled counterclockwise.

- (a) A clockwise rotation with center B and angle θ maps A to C . Find θ such that $\theta < 360^\circ$.
- (b) A clockwise rotation with center B and angle ϕ maps H to D . Find ϕ such that $\phi < 360^\circ$.

16.2.2 B' is the image of B rotated 90° about A . Given $AB = 5$, find BB' .

16.2.3 O is the intersection of the diagonals of quadrilateral $ABCD$. A 90° rotation about O maps $ABCD$ to itself, meaning each vertex of the image is also a vertex of the original quadrilateral. Must $ABCD$ be a square? **Hints:** 43

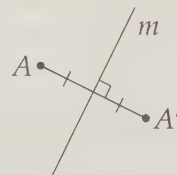
16.2.4★ Regular tetrahedron $ABCD$ is rotated 60° about its altitude from A to face BCD .

- (a) What kind of shape is the intersection of the original tetrahedron and its image? **Hints:** 208
- (b)★ Find the volume of intersection of the original tetrahedron and its image given that $AB = 6$.
Hints: 561

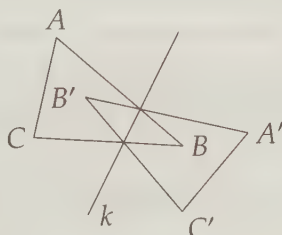
16.3 Reflections

We've tried sliding and we've tried spinning; now, we'll try flipping. If you've looked in a mirror, you have experienced **reflection**.

Mathematically speaking, the image when we **reflect** point A over line m is the point A' such that m is the perpendicular bisector of AA' . In other words, if we folded our paper along line m , A and A' would coincide.



Of course, we can flip more than just points. In the diagram below, $\triangle A'B'C'$ is the image of $\triangle ABC$ upon reflection over line k .



If a figure maps to itself under a reflection over a certain line, that line is called a **line of symmetry** of the figure. For example, every diameter of a circle is part of a line of symmetry of a circle, and each diagonal of a square is part of a line of symmetry of the square.

Problems

Problem 16.6: What are the fixed points of a reflection over a given line m ? In other words, what points are their own images when reflected over line m ?

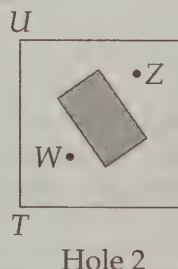
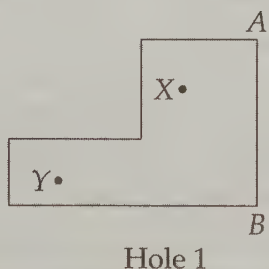
Problem 16.7: Given two different points, X and Y , describe all lines m such that the image of X upon reflection over m is point Y .

Problem 16.8: How many different lines of symmetry does a regular hexagon have?

Problem 16.9: I own a cabin in the woods that is 4 miles directly north of a river that runs east-west. I am currently out exploring the woods. I am 2 miles north of the river, and 3 miles west of my cabin. I want to walk to the river, then walk to the cabin. What is the length of the shortest possible route I can take?

Problem 16.10: My miniature golf course has only two holes, which are shown below.

- On Hole 1, the ball starts at Y , and we want to hit it to X . Find the point on $\overline{AB}$ at which we should aim to hit the ball to X .
- On Hole 2, the ball starts at W , and we want to hit it to Z . There is a barrier between W and Z , as shown. Where should we aim on $\overline{UT}$ to hit the ball to Z ?



One of the best ways to understand any geometric transformation is to analyze what stays the same when everything else is changing.

Problem 16.6: What are the fixed points of a reflection over a given line m ?

Solution for Problem 16.6: Clearly every point on m maps to itself, since m is the 'fold.' To see that there are no other fixed points, we note that m divides the plane into two pieces. Any point on one side of m is mapped to a point on the other side. Since a point can't be on both sides of m , a point not on m can't be mapped to itself! $\square$

Suppose we are given a point and its image under a reflection. How do we find the line over which

the point was reflected?

Problem 16.7: Given two different points, X and Y , describe all lines m such that the image of X upon reflection over m is point Y .

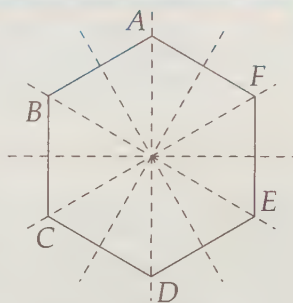
Solution for Problem 16.7: The image of a point A upon reflection over a line is the point A' such that the line is the perpendicular bisector of $\overline{AA'}$. Since the image of X in our problem is Y , the line over which we are reflecting must be the perpendicular bisector of $\overline{XY}$. $\square$

Now that we've mastered reflecting points, let's reflect a more complicated figure.

Problem 16.8: How many different lines of symmetry does a regular hexagon have?

Solution for Problem 16.8: Intuitively, it seems like the only lines of symmetry are the six dashed lines shown that cut the regular hexagon in half, but how can we be sure these are the only ones?

To make sure there aren't any others, we note that if the reflection maps the hexagon to itself, then A must be mapped to some vertex of the hexagon. Furthermore, for any two different vertices, there is only one reflection that will map the vertices to each other – the reflection over the perpendicular bisector of the segment connecting the two vertices. Therefore, there's only one reflection that could possibly map A to B , and one that maps A to C , etc. Each of these reflections gives us a line of symmetry. This doesn't include lines of symmetry through A . $\overleftrightarrow{AD}$ is the only possible line of symmetry through A , since it's the only line through A that leaves two vertices on each side of the line. $\square$



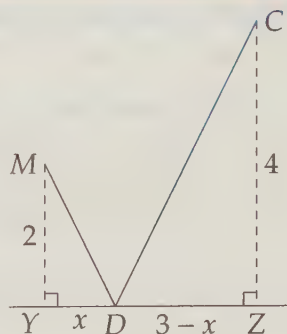
Now we'll use reflections to solve a couple of problems.

Problem 16.9: I own a cabin in the woods that is 4 miles directly north of a river that runs east-west. I am currently out exploring the woods. I am 2 miles north of the river, and 3 miles west of my cabin. I want to walk to the river, then walk to the cabin. What is the length of the shortest possible route I can take?

Solution for Problem 16.9: Point M is where we start, the cabin is at C , and $\overleftrightarrow{YZ}$ is the river. One approach we can take is to suppose we hit the river at point D , x miles east of where we start. We can then use the Pythagorean Theorem to make an expression in terms of x for how far we have to walk. Since $YD = x$, we have $DZ = 3 - x$, so from right triangles $\triangle MYD$ and $\triangle CZD$, we have

$$MD + CD = \sqrt{4 + x^2} + \sqrt{16 + (3 - x)^2}.$$

Figuring out the smallest possible value for this expression is going to be very hard. We need to find a more clever approach.

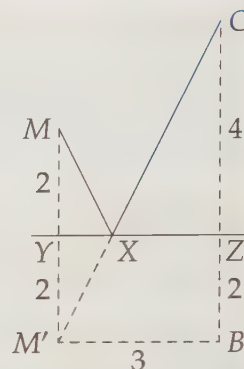


Since we're stuck, we try thinking of similar, simpler problems that we know how to solve. For example, if we didn't have to go to the river, the answer would be easy – we just go from M to C and we're done. Another simplified problem would be if we were on the other side of the river. Then we could still just go straight from our starting point to the cabin, and we'd still hit the river. Aha! We can cross the river.

Suppose we reflect ourselves over the river, to point M' as shown at right. Since $\overline{MX}$ is the mirror image of $\overline{M'X}$, we have $MX + XC = M'X + XC$. Therefore, finding the shortest total distance from M to the river to C is the same as finding the shortest total distance from M' to the river to C .

Any path from M' to C hits the river, so clearly our desired shortest distance is $M'C$. Since M is 2 miles north of the river, M' is 2 miles south of it. So, M' is $2 + 4 = 6$ miles south of the cabin. Since M' is also 3 miles west of the cabin, $M'C = \sqrt{36 + 9} = 3\sqrt{5}$.

Therefore, the shortest path I can take is $3\sqrt{5}$ miles long. $\square$

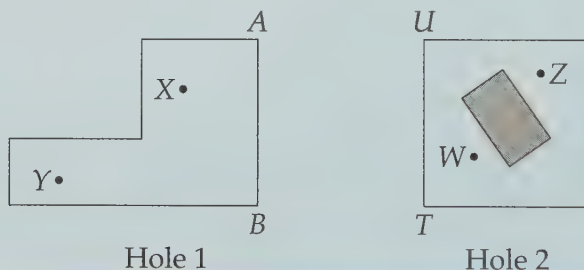


Concept: If you can't solve a problem, try thinking of a similar, simpler problem and solve that. Then try to use your solution to the simpler problem to solve the harder problem.

Next time you play miniature golf or shoot pool, you'll likely think of reflections. Here's why:

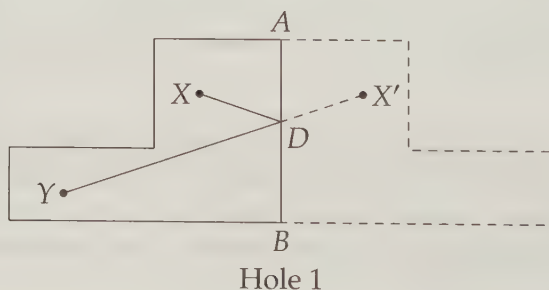
Problem 16.10: My miniature golf course has only two holes, as shown below.

- On Hole 1, the ball starts at Y , and we want to hit it to X . Find the point on $\overline{AB}$ at which we should aim to hit the ball to X .
- On Hole 2, the ball starts at W , and we want to hit it to Z . There is a barrier between W and Z , as shown. Where should we aim on $\overline{TU}$ to hit the ball to Z ?



Solution for Problem 16.10:

- For our first hole, we need to bounce the ball off $\overline{AB}$ so that it goes directly to X . To figure out where to aim the ball, we note that the path after the ball bounces is the reflection of what the path would be if there were no wall $\overline{AB}$ and the ball just kept going straight. Therefore, if we aim at the image of X upon reflection over $\overline{AB}$, the ball will bounce off $\overline{AB}$ and go straight to X .



So, to find the point on $\overline{AB}$ to aim at, we reflect X over $\overline{AB}$ to get its image X' . The intersection of $\overline{YX'}$ and $\overline{AB}$ is the point we should target.

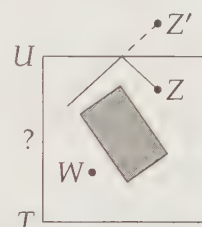
- (b) On Hole 2, the barrier not only prevents us from hitting the ball straight from W to Z , but it also prevents us from just bouncing the ball off $\overline{TU}$ and having it go straight to Z . We'll have to go for a double-bank shot. Therefore, we need it to bounce off $\overline{TU}$, then hit the appropriate point on the top wall so that it continues to point Z .

We still don't know where to aim along $\overline{TU}$, but we can figure out where we want the ball heading after it banks off of $\overline{TU}$. After coming off $\overline{TU}$, the ball should be heading towards the image of Z upon reflection over the top wall.

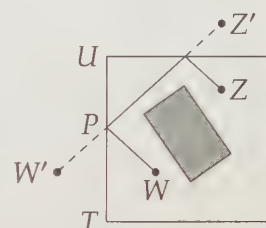
We now know where we want the ball heading once it comes off $\overline{TU}$, but we still don't know where to aim along $\overline{TU}$. Since we need the ball to go from W and bounce off $\overline{TU}$ so that it is heading straight towards Z' , we think to try our reflection tactic on W as well.

Point W' is the image of W upon reflection over $\overline{TU}$. If we were at W' and there were no wall $\overline{TU}$, we would aim directly at Z' . The path $\overline{W'Z'}$ meets $\overline{TU}$ at P . Since $\overline{W'P}$ is the image of $\overline{WP}$, if we aim the ball at P , it will bounce off $\overline{TU}$, and head towards Z' . Since Z' is the image of Z upon reflection over the top wall, a ball heading from P towards Z' will bounce off the top wall and head straight to Z as desired.

Have fun beating your friends next time you play miniature golf!



Hole 2



Hole 2

Exercises

16.3.1 How many lines of symmetry does each of the following figures have?

- A rectangle that is not a square.
- A rhombus that is not a rectangle.
- A parallelogram that is not a rectangle or a rhombus.
- A regular pentagon.
- A regular dodecagon.
- A regular polygon with n sides.

16.3.2 Square $ABCD$ is reflected over $\overleftrightarrow{CD}$. A' and B' are the images of A and B , respectively. Given $CD = 4$, find $A'A$ and $A'B$.

16.3.3 Lines k and ℓ are not parallel, and line m is the image of line k upon reflection over ℓ . Show that ℓ bisects a pair of the angles formed by line m and k .

16.3.4 Given intersecting lines ℓ and m and point X not on either line, let Y be the image of X upon reflection over ℓ and let Z be the image of X upon reflection over m .

- Is it possible for the reflection of Y over m and the reflection of Z over ℓ to be the same point?
- Must the reflection of Y over m and the reflection of Z over ℓ be the same point?

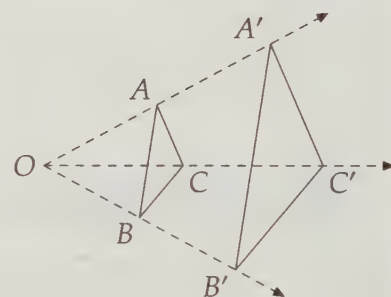
16.3.5★ We can define reflections in space through a plane similar to the way we define reflections over a line. Specifically, given a point K and a plane $\mathcal{P}$, then the image of K upon reflection through $\mathcal{P}$ is the point K' such that $\mathcal{P}$ is perpendicular to $\overline{KK'}$ and bisects $\overline{KK'}$. A **plane of symmetry** of a figure is a plane such that the image upon reflecting the figure through the plane is the figure itself. For example, any plane through the center of a sphere is a plane of symmetry of the sphere.

- How many planes of symmetry does a right square pyramid have?
- How many planes of symmetry does a right rectangular prism have if the dimensions of the prism are all different?
- How many planes of symmetry does a regular tetrahedron have? **Hints:** 154
- How many planes of symmetry does a cube have? **Hints:** 63
- Show that any plane of symmetry of a right circular cone must include the axis of the cone. **Hints:** 115

16.4 Dilation

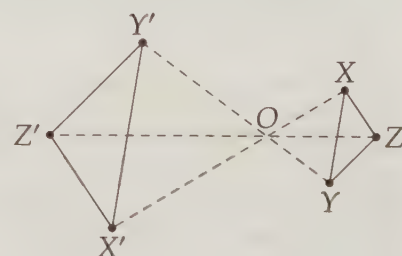
The last transformation we will explore is **dilation**. In plain English, a dilation results from stretching or shrinking a figure.

At right is an example of dilation. Point O is the **center of dilation**. The image of point A is the point A' on $\overrightarrow{OA}$ such that $OA'/OA = 2$. Similarly, the image of point B is the point B' on $\overrightarrow{OB}$ such that $OB'/OB = 2$, and the image of point C is the point C' on $\overrightarrow{OC}$ such that $OC'/OC = 2$. The image of the center of dilation is itself. We can think of the dilation as 'stretching' $\triangle ABC$ away from O .



In general, the image of a point P upon dilation with **scale factor** k and center O is the point P' on $\overrightarrow{OP}$ such that $OP' = k(OP)$ (when k is positive). In our example above, the scale factor is 2.

The scale factor need not be positive. At right we have an example in which the scale factor is -2 . Here, the image of point X upon dilation about O is the point X' on the ray starting from O but going in the opposite direction from X such that $OX'/OX = 2$. In other words, point X' is the point such that $OX'/OX = 2$ and O is on $\overline{XX'}$.



Problems

Problem 16.11: Triangle $A'B'C'$ is the image of a dilation of triangle ABC about the centroid of triangle ABC . The dilation has a positive scale factor.

- Suppose $\triangle A'B'C'$ is entirely outside $\triangle ABC$. Is the scale factor greater than 1 or less than 1?
- Suppose $\triangle A'B'C'$ is entirely inside $\triangle ABC$. Is the scale factor greater than 1 or less than 1?
- What happens if the scale factor equals 1?

Problem 16.12: Square $A'B'C'D'$ is the image of a dilation of square $ABCD$ about A with scale factor 3. Suppose $AB = 2$.

- Find AA' , AB' , and BB' .
- Find BC' and CD' .

Problem 16.13: Let A' and B' be the images of points A and B , respectively, under a dilation with center O and positive scale factor k .

- Show that $\triangle AOB \sim \triangle A'O'B'$.
- Show that $\overline{AB} \parallel \overline{A'B'}$ and $A'B'/AB = k$.

Problem 16.11: Triangle $A'B'C'$ is the image of a dilation of $\triangle ABC$ about the centroid of triangle $\triangle ABC$.

- Suppose $\triangle A'B'C'$ is entirely outside $\triangle ABC$. Is the scale factor greater than 1 or less than 1?
- Suppose $\triangle A'B'C'$ is entirely inside $\triangle ABC$. Is the scale factor greater than 1 or less than 1?
- What happens if the scale factor equals 1?

Solution for Problem 16.11: Let G be the centroid of $\triangle ABC$. The centroid of a triangle is always inside the triangle, so G is inside $\triangle ABC$.

- Because A' is the image of A upon dilation about G , point A' is on $\overrightarrow{GA}$. If A' is outside the triangle on this ray, then it is beyond A on $\overrightarrow{GA}$. So, we must have $GA' > GA$, which means $GA'/GA > 1$. Therefore, the scale factor is greater than 1. An example of this case is shown at left below.

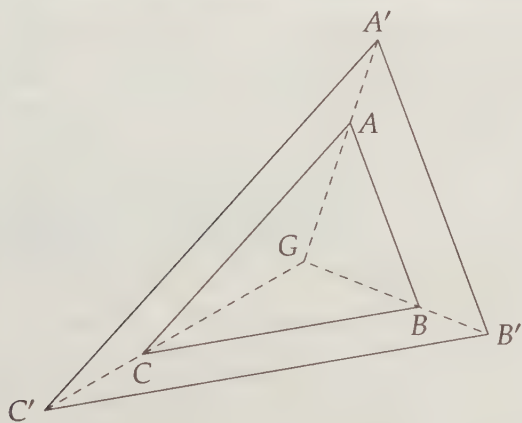


Figure 16.2: Scale Factor Greater Than 1

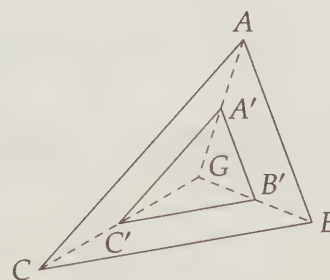


Figure 16.3: Scale Factor Less Than 1

- As before, we know that A' is on $\overrightarrow{GA}$. However, this time, A' is inside $\triangle ABC$, so we have $GA' < GA$. Therefore, we have $GA'/GA < 1$, so the scale factor is less than 1. An example of this case is shown at right above.
- If the scale factor equals 1, then the image of $\triangle ABC$ is itself.

□

Important:


If the scale factor of a dilation is greater than 1, then the dilation of a figure 'stretches' the figure. The resulting image is larger than the original figure. If the scale factor is positive but less than 1, then the dilation is a 'shrinking' of the original figure, which gives an image that is smaller than the original figure. A dilation of a figure with scale factor equal to 1 leaves the figure unchanged.

Problem 16.12: Square $A'B'C'D'$ is the image of a dilation of square $ABCD$ about A with scale factor 3. Suppose $AB = 2$.

- Find AA' , AB' , and BB' .
- Find BC' and CD' .

Solution for Problem 16.12:

- At right, we have the original square $ABCD$ and its image under the dilation with scale factor 3 and center A . The image of A upon a dilation about A is just itself. (The image of the center of a dilation is always the center itself.) Therefore, we have $AA' = 0$. (Yeah, that was a bit of a trick question!)

Because B' is the image of B upon a dilation about A with scale factor 3, we have $AB'/AB = 3$. In other words, B' is 3 times as far from A as B is. Since $AB = 2$, we have $AB' = 6$. Points A , B , and B' are collinear because B' is the image of B upon the dilation about A . So, we have $AB + BB' = AB'$. From this, we find that $BB' = AB' - AB = 4$.

- We seek two lengths and we have lots of right angles, so we look for right triangles on which we can use the Pythagorean Theorem. In the previous part, we found that $AB' = 6$, so square $AB'C'D'$ has side length 6. We also found that $BB' = 4$, so right triangle $BB'C'$ gives us

$$BC' = \sqrt{(BB')^2 + (CC')^2} = \sqrt{16 + 36} = 2\sqrt{13}.$$

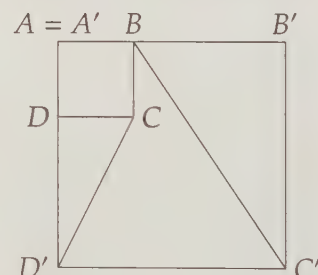
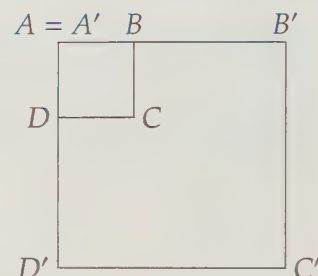
In the same way we found $BB' = 4$ in part (a), we can find that $DD' = 4$. Right triangle CDD' then gives us

$$CD' = \sqrt{CD^2 + D'D^2} = \sqrt{4 + 16} = 2\sqrt{5}.$$

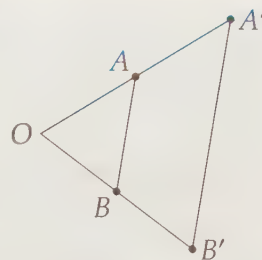
□

This problem raises an interesting question about dilations. Points B' and C' are the images of B and C under the dilation, and we see that $\overline{B'C'} \parallel \overline{BC}$, and $B'C'/BC$ equals the scale factor. Note that we didn't prove these facts; we were told to assume that the image of $ABCD$ upon dilation is a square. Let's see if we can prove that these observations are not a coincidence.

Problem 16.13: Let A' and B' be the images of points A and B , respectively, under a dilation with center O and positive scale factor k . Show that $\overline{AB} \parallel \overline{A'B'}$ and $A'B'/AB = k$.



Solution for Problem 16.13: We want to prove something about a ratio, and we want to show that two lines are parallel. The dilation also gives us information about ratios. This looks like a job for similar triangles. Our diagram at right shows the set-up where $k > 1$. (The case $k < 1$ can be solved with the same steps we will use for $k > 1$.)



We're looking for similar triangles, and we need to find some way to use the fact that A' and B' are the images of A and B under the same dilation about O . The dilation information tells us that $\overrightarrow{OA}$ passes through A' and $\overrightarrow{OB}$ passes through B' , as shown. But that's not enough! The dilation also gives us information about ratios. Specifically, it tells us that OA'/OA and OB'/OB both equal the scale factor, so

$$\frac{OA'}{OA} = \frac{OB'}{OB} = k.$$

Since we also have $\angle AOB = \angle A'OB'$, we have $\triangle AOB \sim \triangle A'OB'$ by SAS Similarity. From this triangle similarity, we have $A'B'/AB = OA'/OA = k$. We also have $\angle OAB = \angle OA'B'$, which tells us that $\overline{AB} \parallel \overline{A'B'}$.

□

Notice that we did not prove in Problem 16.13 that the image of $\overline{AB}$ is $\overline{A'B'}$. We only proved that $A'B'/AB = k$ and $\overline{AB} \parallel \overline{A'B'}$. You'll have a chance to prove that the image of a segment under dilation is a segment as a challenging Exercise. However, the parallel lines suggest something very useful about dilations.

Important:

A figure and its image upon dilation are similar. The ratio of corresponding sides of the figure and its image equals the scale factor of the dilation.

You'll have a chance to prove this for triangles as an Exercise. This is just the beginning of interesting dilation properties. In Art of Problem Solving's *Intermediate Geometry*, we will build on these basics to find powerful and intriguing applications of dilation.

Exercises

16.4.1 In $\triangle XYZ$, we have $XY = 3$, $YZ = 4$, and $XZ = 5$. Suppose $\triangle X'Y'Z'$ is the image of $\triangle XYZ$ under a dilation with scale factor 4.

- (a) What is $X'Y'$?
- (b) What is the area of $\triangle X'Y'Z'$?
- (c) Is it possible to determine XX' with the information given in the problem?

16.4.2 Point Q is the image of point P under a dilation with center O and scale factor 5. If $PQ = 20$, what is OP ?

16.4.3 Show that if A' , B' , and C' are the images of A , B , and C , respectively, under a dilation with center O and scale factor k , then $\triangle ABC \sim \triangle A'B'C'$. (You can assume k is positive.) **Hints:** 6

16.4.4★ In this problem, we show that the image of a segment upon a dilation is itself a segment. Suppose that A' and B' are the images of A and B , respectively, upon dilation about O with scale factor k . (You can assume $k > 0$; the proof for $k < 0$ is very similar.)

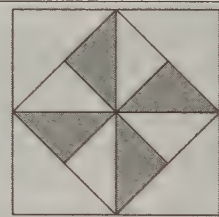
- Let X be a point on $\overline{AB}$, and let X' be the image of X under the dilation. Show that $\angle OXA = \angle OX'A'$ and $\angle OXB = \angle OX'B'$.
- Use part (a) to show that $\angle A'X'B' = 180^\circ$. Why does this mean that the image of each point on $\overline{AB}$ is a point on $\overline{A'B'}$?
- Show that each point on $\overline{A'B'}$ is the image of some point on $\overline{AB}$ under the dilation. Why is this step necessary to complete the proof?

16.5 Changing the Question

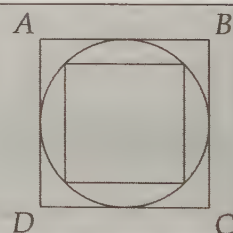
Sometimes problems can be quickly solved by rearranging the problem or by considering special cases of the problem. Typically this requires a little creativity and flexibility.

Problems

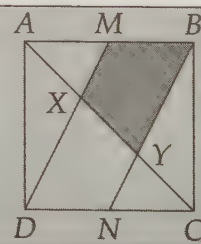
Problem 16.14: The side length of the largest square at right is 10 inches. The midpoints of the sides of the largest square are connected to form a smaller square. Opposite vertices and the midpoints of opposite sides of the smaller square are then connected as shown. What is the number of square inches in the area of the shaded region? (Source: MATHCOUNTS)



Problem 16.15: A circle is inscribed in a large square and circumscribed about a smaller square. The area of the larger square is 8 square meters. What is the number of square meters in the area of the smaller square? (Source: MATHCOUNTS)



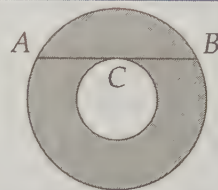
Problem 16.16: In the figure, $ABCD$ is a square with side length 1, and M and N are the midpoints of $\overline{AB}$ and $\overline{CD}$, respectively. Find the area of the shaded region. (Source: Mandelbrot)



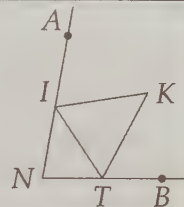
Extra! Every new body of discovery is mathematical in form, because there is no other guidance we can have.

—Charles Darwin

Problem 16.17: In the diagram we have two concentric circles, and chord $\overline{AB}$ of the large circle is tangent to the smaller circle. Given that $AB = 8$, find the area of the region between the two circles.

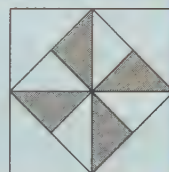


Problem 16.18: In the diagram, $\angle INT = 80^\circ$. The angle bisectors of two of the exterior angles of $\triangle TIN$ meet at point K , as shown. Compute $\angle IKT$.



Since rotations, reflections, and translations don't alter lengths or areas, we can often use them to simplify problems. Here are several examples.

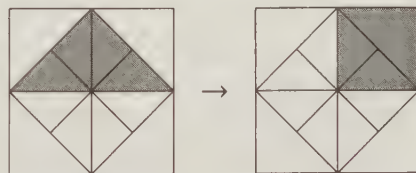
Problem 16.14: The side length of the largest square at right is 10 inches. The midpoints of the sides of the largest square are connected to form a smaller square. Opposite vertices and the midpoints of opposite sides of the smaller square are then connected as shown. What is the number of square inches in the area of the shaded region? (Source: MATHCOUNTS)



Solution for Problem 16.14: Solution 1: Each of the shaded regions is a 45-45-90 triangle, so we can find the area of each by finding one of the legs. One leg of each triangle is half a diagonal of one of the 5×5 quarters of the largest square. Therefore, each triangle has a leg of length $5\sqrt{2}/2$, so each triangle has an area of $(5\sqrt{2}/2)^2/2 = 25/4$. There are four such triangles, so the total shaded area is 25.

Solution 2: Instead of finding the leg lengths of each little triangle to find the area, we could have instead noted that each of these little triangles is $1/4$ of one of the 5×5 squares. Hence, each has area $25/4$, so the total shaded area is $4(25/4) = 25$.

Solution 3: Since the area of a triangle doesn't change if we slide, spin, or flip it, we can manipulate the shaded pieces to make a region whose area we know how to find easily. We first slide the bottom triangles up to complete a large shaded isosceles right triangle. Then, we can slide half of this triangle to the right to complete one of the 5×5 squares as shown. Clearly, then, the area of the shaded region is $5^2 = 25$. $\square$

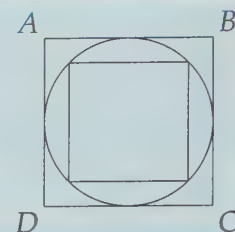


Concept: In addition to our strategy of chopping areas into easy-to-handle pieces, we can also sometimes rearrange them to form an easy-to-handle whole.



Extra! **Fagnano's Problem** asks what is the triangle of smallest perimeter that can be formed by connecting a point on each side of an acute triangle. The answer and a dazzling solution is on page 425.

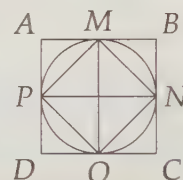
Problem 16.15: A circle is inscribed in a large square and circumscribed about a smaller square. The area of the larger square is 8 square meters. What is the number of square meters in the area of the smaller square?



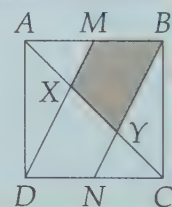
Solution for Problem 16.15: As with the last problem, we can work through several steps to the answer, or we can make a little transformation to simplify the problem.

Solution 1: Since our large square has area 8, each side has length $\sqrt{8} = 2\sqrt{2}$. Because the circle is tangent to all four sides of the large square, the diameter of the circle equals the side length of the large square. But the diagonal of the small square is also a diameter of the circle, so the diagonal of the small square has length $2\sqrt{2}$. Therefore, its sides are each $(2\sqrt{2})/\sqrt{2} = 2$ and its area is 4.

Solution 2: We can spin the small square without changing its area, so we spin it so that its vertices are the points where the circle is tangent to the large square. When we draw the diagonals of the small square $MNOP$, we divide the large square into four equal pieces. Half of each of these pieces is inside $MNOP$, so $[MNOP] = [ABCD]/2 = 4$.



Problem 16.16: In the figure, $ABCD$ is a square with side length 1, and M and N are the midpoints of $\overline{AB}$ and $\overline{CD}$, respectively. Find the area of the shaded region. (Source: Mandelbrot)

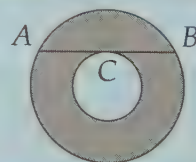


Solution for Problem 16.16: $MBYX$ is a trapezoid (why?), but it's not clear how we'll find the height or either of the bases. It's also not clear how we could chop $MBYX$ into easy-to-handle pieces. However, it looks like $MBYX$ is congruent to $NDXY$. We don't have any tools for proving quadrilaterals are congruent like we do for triangles. But if we can transform one quadrilateral into the other with flips, spins, or slides, then we'll know they are congruent.

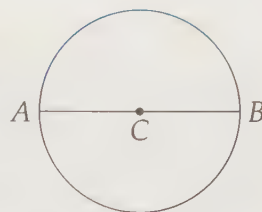
We might think to reflect $\triangle ABC$ over $\overline{AC}$, but that won't map M to N . However, if we rotate $\triangle ABC$ 180° about the center of the square, we'll have the transformation we want. (You'll prove this as an Exercise.) This rotation maps $MBYX$ to $NDXY$, showing that these quadrilaterals are indeed congruent. Therefore, the area of $MBYX$ is half the area of $MBND$. $MBND$ is a parallelogram with base $ND = 1/2$ and height $BC = 1$, so its area is $1/2$. Therefore, the shaded region has area $(1/2)/2 = 1/4$. $\square$

Transformations aren't the only ways we can manipulate diagrams to help us with problems.

Problem 16.17: In the diagram we have two concentric circles, and chord $\overline{AB}$ of the large circle is tangent to the smaller circle. Given that $AB = 8$, find the area of the region between the two circles.

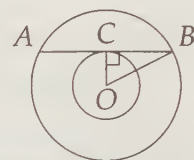


Solution for Problem 16.17: The problem doesn't tell us what the radius of either the larger circle or the smaller circle is. We only know the length of that one tangent chord. Therefore, the problem suggests that the shaded region between the circles has the same area *no matter what these two radii are*, as long as that tangent chord has length 8. We can get a quick answer for what that area is by picking a special case that's easy to work with. The simplest is to shrink the smaller circle down to a point. Then, our tangent chord becomes a diameter of the larger circle, and our shaded region is this whole circle. The area of this region, then, is just $(8/2)^2\pi = 16\pi$.



Notice that we have **not** proved that the shaded region is always 16π when our tangent chord is 8 units long. We've only found it for one case. However, if the area is always the same, we know it's 16π . To prove that it's always the same, we have to find some way to get information about the radii of the two circles. We have a tangent line, so naturally we build a right triangle by drawing radii as shown below.

Since $\overline{OC} \perp \overline{AB}$, $\overline{OC}$ bisects chord $\overline{AB}$. Therefore, $BC = 4$. From right triangle $\triangle OBC$, we have $OB^2 - OC^2 = BC^2 = 16$. Since $\overline{OB}$ and $\overline{OC}$ are the radii of our circles, the area we seek is $(OB^2)\pi - (OC^2)\pi = (OB^2 - OC^2)\pi = 16\pi$. $\square$



Concept: When working on a problem with a variable set-up, consider the extreme possibilities.

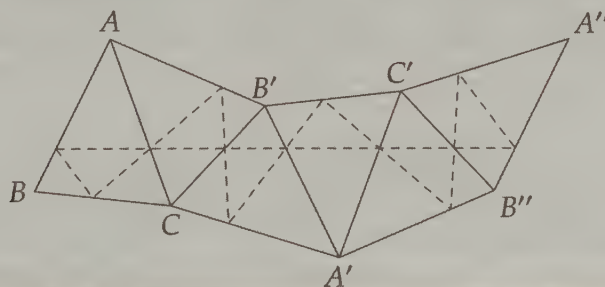


WARNING!!



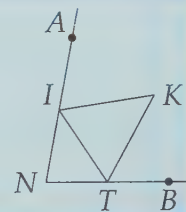
Before you start examining those extremes, make sure you really can vary the set-up. If you vary the set-up in a way that violates information given in the problem, then your conclusions don't necessarily apply to the original problem! For example, if we are asked to find the area of a triangle with sides of length 13, 14, and 15, we can't simply pretend two of the sides are perpendicular to get the answer!

Extra! In an acute triangle, the triangle formed by connecting the feet of the altitudes is the triangle with smallest perimeter that can be formed by connecting a point on each side of the triangle. This was originally proved by Fagnano using calculus, but mathematician HA Schwarz came up with the amazing proof without words offered below, using the property of reflection repeatedly. See if you can figure out how this proof works!



Let's try this technique on one more problem.

Problem 16.18: In the diagram, $\angle INT = 80^\circ$. The angle bisectors of two of the exterior angles of $\triangle TIN$ meet at point K , as shown. Compute $\angle IKT$.



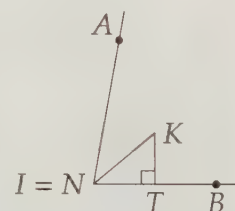
Solution for Problem 16.18: Since $\angle ITN$ can vary, we'll call it x . From $\triangle INT$, we find that $\angle TIN = 180^\circ - 80^\circ - x = 100^\circ - x$. Therefore, $\angle AIT = 180^\circ - \angle TIN = 80^\circ + x$ and $\angle ITB = 180^\circ - x$. Since $\overline{IK}$ and $\overline{TK}$ bisect these angles, we have $\angle KIT = 40^\circ + x/2$ and $\angle KTI = 90^\circ - x/2$. Finally, we can use $\triangle KIT$:

$$\angle K + \angle KIT + \angle KTI = 180^\circ.$$

Substitution gives $\angle K + (40^\circ + x/2) + (90^\circ - x/2) = 180^\circ$, from which we find $\angle K = 50^\circ$.

Notice that in this problem, we can place points I and T anywhere on the sides of $\angle ANB$ without violating the problem. We only need $\angle INT = 80^\circ$. Therefore, to quickly check the answer we found above, we can consider the extreme case of putting I at N .

When I is at N , $\angle AIT$ is $\angle N$ and $\overline{IK}$ is the bisector of $\angle N$. Therefore, $\angle KIT = 80^\circ/2 = 40^\circ$. Furthermore, $\angle ITB$ is a straight angle, so its bisector is perpendicular to $\overline{IT}$. Since $\triangle KIT$ is a right triangle, we then have $\angle IKT = 90^\circ - 40^\circ = 50^\circ$. Notice that this is not a proof that $\angle IKT$ is always 50° – it's just a quick way to check the answer we found earlier. $\square$

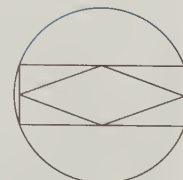


Concept: Considering extremes in problems with varying set-ups is a great way to check your answer. But remember this warning:

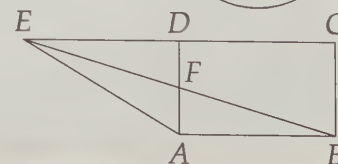
WARNING!! Considering extremes does not constitute a proof! You still have to show your answer holds in all cases.

Exercises

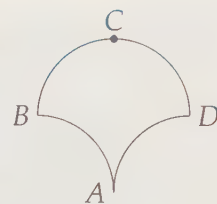
16.5.1 The rectangle at right is inscribed in the circle, and a rhombus is inscribed in the rectangle by connecting the midpoints of the sides of the rectangle as shown. If the radius of the circle is 4 ft, how many feet are in the perimeter of the rhombus? (Source: MATHCOUNTS)



16.5.2 In the diagram, D is the midpoint of $\overline{EC}$, and the area of $\triangle EDF$ is 4 square centimeters. What is the number of square centimeters in the area of rectangle $ABCD$? (Source: MATHCOUNTS)



16.5.3 Three circular arcs of radius 5 units bound the region shown. Arcs $\widehat{AB}$ and $\widehat{AD}$ are quarter-circles, and $\widehat{BCD}$ is a semicircle. What is the area, in square units, of the region?



16.5.4 In Problem 16.16, we asserted that $DNYX$ is the image of $BMXY$ upon rotation about the center of the square. Complete our solution by proving this fact. **Hints:** 419

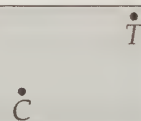
16.5.5★ In trapezoid $ABCD$, $\overline{AB}$ and $\overline{CD}$ are perpendicular to $\overline{AD}$, with $AB + CD = BC$ and $AD = 7$. What is $AB \cdot CD$? (Source: AMC 10) **Hints:** 497

16.6 Construction: Transformations

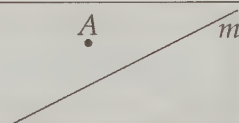
As a thorough test of your spinning and flipping abilities, you'll now have the chance to spin and flip some points given only a ruler and compass.

Problems

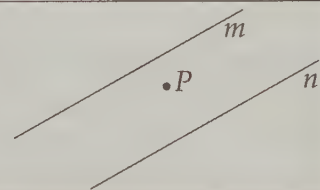
Problem 16.19: Given are points C and T . Construct point U such that U is the image of point T under a 30° rotation about C .



Problem 16.20: Given is point A and line m . Construct point B such that B is the image of point A upon reflection over m .



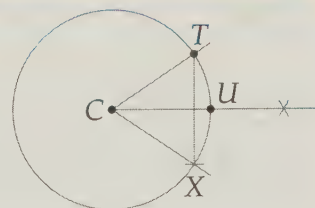
Problem 16.21: Given are parallel lines m and n and point P between the lines. Construct an equilateral triangle such that one vertex is P , one vertex is on m , and one vertex is on n .



Let's go for a little spin.

Problem 16.19: Given are points C and T . Construct point U such that U is the image of point T under a 30° rotation about C .

Solution for Problem 16.19: Since U is the image of T rotated 30° about C , we must have $CT = CU$ and $\angle TCU = 30^\circ$. Since $CT = CU$, U must be just as far from C as T is. Therefore, U must be on the circle with center C and radius CT . So, we draw that circle. Since $\angle TCU = 30^\circ$, U must be on a ray through C that forms a 30° angle with $\overline{CT}$. We construct this ray by first constructing an equilateral triangle, $\triangle CTX$, then bisecting $\angle C$ of this triangle.



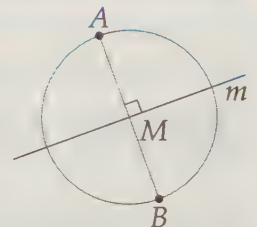
The intersection of this angle bisector and our circle is the desired point U such that $CT = CU$ and $\angle TCU = 30^\circ$. Therefore, U is the image of T under a 30° rotation about C . $\square$

Concept: In construction problems in which we must locate a single point, our general strategy is to find two figures, usually circles or lines, which must contain the point. The intersection of these two figures is the point we want.

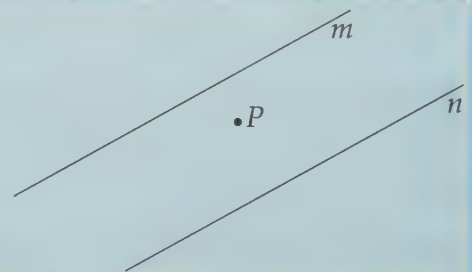
You'll flip over this one...

Problem 16.20: Given is point A and line m . Construct point B such that B is the image of point A upon reflection over m .

Solution for Problem 16.20: Since B is the image of point A upon reflection over m , we must find the point B such that m is the perpendicular bisector of $\overline{AB}$. Since $\overline{AB} \perp m$, B must be on the line through A perpendicular to m (see Problem 6.24 for this construction). Let this perpendicular line meet m at M . Since m bisects $\overline{AB}$, we must have $AM = BM$. Therefore, B is on the circle with center M and radius AM . Where this circle meets our earlier line through A is the desired point B such that m is the perpendicular bisector of $\overline{AB}$. $\square$



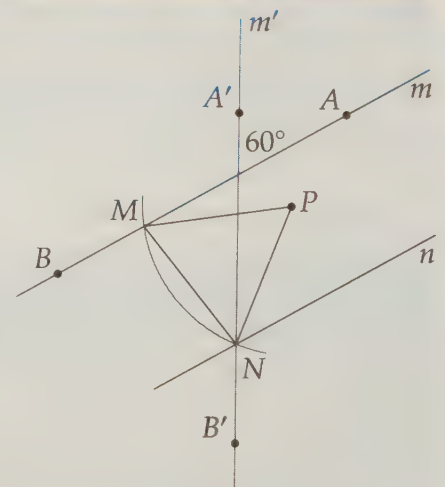
Problem 16.21: Given are parallel lines m and n and point P between the lines. Construct an equilateral triangle such that one vertex is P , one vertex is on m , and one vertex is on n .



Solution for Problem 16.21: We seek point M on m and point N on n such that $\triangle PMN$ is equilateral. If we can find either M or N , we can easily construct our equilateral triangle, since once we have a side of the triangle, we can make the whole thing. Therefore, we'll focus on finding N .

We already have one figure that contains N – line n . Therefore, we only need to find one more line or circle that contains N . Point P and line n are not enough to find N ; we need more. The only other piece of information we have is that M is on m , so we need to use m to find N .

Since $\triangle PNM$ must be equilateral, we know that a 60° rotation about P maps M to N . But we don't know where M is! However, we do know that M is on m , so the image of M upon rotating 60° about P must be on the image of m when m is rotated 60° about P . Since this image of point M is point N , all we have to do is rotate m 60° about point P to have another line that contains N !



To rotate m 60° about P , we simply pick two points on m and rotate them both. Connecting the images of these points upon rotation gives us m' , the image of m upon 60° rotation about P . Since N is on both m' and n , where these lines meet gives us N ! Now we can easily use $\overline{PN}$ to construct the equilateral triangle. $\square$

One key step in finding our solution was thinking about what information we hadn't used yet, thereby focusing on using line m to find N . When we focused on how to use m to get N , we had to think about how to relate M to N given point P . This led us to a new problem solving tactic:

Concept: When stuck on a problem involving equilateral triangles, think about 60° rotations!



Exercises

16.6.1 Given points A , A' , and Q , construct point Q' such that Q' is the image of Q under the same translation that maps A to A' .

16.6.2 Given a point P and two intersecting lines, ℓ and m , construct the image of P upon reflection over ℓ . Let P' be the image of P upon reflection over ℓ . Construct the image of P' upon reflection over m . Try this for a variety of points (reflecting the point over ℓ , then reflecting its image over m). Notice anything interesting?

16.7 Summary

Definitions: In this chapter we explored three important geometric transformations. A figure's **image** under a transformation is the result of applying the transformation to that figure. We also say that a transformation **maps** a figure to its image. A **fixed point** of a transformation is a point that is its own image. Four important transformations are described below.

- When we slide a figure a given amount in a specified direction, we perform a **translation**.
- When we apply a **rotation** to a figure, we spin it by some angle about some point. The point is the **center** of rotation.
- The image of a figure upon **reflection** over a line is what results when the figure is flipped over the line. If a figure is its own image upon reflection over a line, the line is a **line of symmetry** of the figure.
- Loosely speaking, when we apply a **dilation** about a point O to a figure, we stretch the figure away from O or shrink it towards O .

Translations, rotations, and reflections preserve length, area, and angles, which means that the lengths, areas, and angles in a figure's image equal the corresponding quantities in the figure.

Problem Solving Strategies

Concepts:



- Understanding transformations is often more about learning what stays the same rather than what changes. Whenever you are learning about a new transformation, focus on what stays the same.
- If you can't solve a problem, try thinking of a similar, simpler problem and solve that. Then try to use your solution to the simpler problem to solve the harder problem.
- In addition to our strategy of chopping areas into easy-to-handle pieces, we can also sometimes rearrange them to form an easy-to-handle whole.
- When working on a problem with a variable set-up, consider the extreme possibilities. Considering extremes is also a great way to check your answer.
- In construction problems in which we must locate a single point, our general strategy is to find two figures, usually circles or lines, which must contain the point. The intersection of these two figures is the point we want.
- When stuck on a problem involving equilateral triangles, think about 60° rotations!

Things To Watch Out For!

WARNING!!



- Before you start examining extremes in a problem, make sure you really can vary the set-up. If you vary the set-up in a way that violates information given in the problem, then your conclusions don't necessarily apply to the original problem!
- Considering extremes does not constitute a proof! You still have to show your answer holds in all cases.

Extra! Suppose we have a circle $\odot O$ with radius r . The geometric transformation called **inversion** maps each point P other than O to the point P' on $\overrightarrow{OP}$ such that $OP \cdot OP' = r^2$. To make inversion work, we define the **point at infinity** to be the image of the center of inversion, O . Experiment with inversion yourself by finding out what happens to various points upon inversion. What happens to points inside the circle? Points outside the circle? Points on the circle? After playing with inversion for a while, turn to page 434 for more suggestions for exploring inversion.

REVIEW PROBLEMS

16.22 $ABCDEFGH$ is a regular octagon. A translation maps G to D . To what point does this same translation map H ?

16.23 $OPQRSTUVWXYZ$ is a regular dodecagon.

- (a) A rotation of θ_1 degrees about U maps V to T . Given $\theta_1 < 180^\circ$, find θ_1 .
- (b) A rotation of θ_2 degrees about U maps X to R . Given $\theta_2 < 180^\circ$, find θ_2 .
- (c) A rotation of θ_3 degrees about U maps W to S . Given $\theta_3 < 180^\circ$, find θ_3 .
- (d) A rotation of θ_4 degrees about U maps Y to Q . Given $\theta_4 < 180^\circ$, find θ_4 .
- (e) A rotation of θ_5 degrees about U maps Z to P . Given $\theta_5 < 180^\circ$, find θ_5 .

16.24 All of the angles of octagon $GHIJKLMN$ are 135° . $GH = IJ = KL = MN = 1$ and $HI = JK = LM = GN = 2$. How many lines of symmetry does $GHIJKLMN$ have?

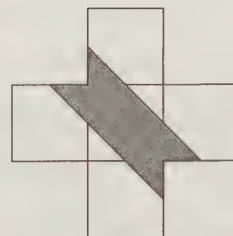
16.25

- (a) $\overline{GH}$ and $\overline{GI}$ are line segments such that $GH = GI$. Must there exist a reflection that maps $\overline{GH}$ to $\overline{GI}$?
- (b) $\overline{XY}$ and $\overline{AB}$ are line segments such that $XY = AB$. Must there exist a reflection that maps $\overline{XY}$ to $\overline{AB}$?

16.26 $ABCD$ is a square. M is the midpoint of $\overline{CD}$.

- (a) Is there a rotation about M that maps A to B ?
- (b) Is there a rotation about M that maps A to C ?
- (c) Is there a reflection over some line that maps M to A ?

16.27 In the figure at right, the shaded region is formed by drawing two parallel segments that connect the midpoints of congruent squares. Each square has side length 1 centimeter. What is the area of the shaded region? (Source: MATH-COUNTS)



16.28 The image upon reflection of regular hexagon $QWERTY$ over $\overleftrightarrow{WE}$ is regular hexagon $Q'W'E'R'T'Y'$.

- (a) Why must W' and E' be W and E , respectively?
- (b) Find $\angle QWQ'$.
- (c)★ Find YT' if $QW = 6$. **Hints:** 435

16.29 Line ℓ is a line of symmetry of $\triangle ABC$.

- (a) Must line ℓ pass through a vertex of $\triangle ABC$? Why or why not?
- (b) Is it possible for $\triangle ABC$ to be scalene (i.e., all sides have different lengths)? Why or why not?

16.30

- (a) Point R is the image of point S under a dilation with center T and scale factor 3. If $TR = 18$, what is TS ?
- (b) Point E is the image of point D under a dilation with center O and scale factor -3 . If $ED = 9$, what is OD ?
- (c) Point X is the image of point Y under a dilation with center A and scale factor 0.25. If $XY = 20$, what is AY ?

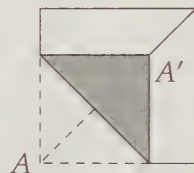
16.31 The image of circle $\odot O$ under a dilation about P with scale factor 4 is $\odot O'$. The area of $\odot O'$ is 48π . What is the area of $\odot O$?

16.32 The rules for a race require that all runners start at A , touch any part of the 1200-meter wall, and stop at B . What is the number of meters in the minimum distance a participant must run?



16.33 Lines ℓ and m are perpendicular at point X . Point A is not on either line. B is the image of A upon reflection over ℓ , and C is the image of B upon reflection over m . Show that $\angle ABC = 90^\circ$ and $XA = XB = XC$. Must $\overline{AC}$ pass through X ?

16.34 A square sheet of paper has area 6 cm^2 . The front is white and the back is shaded. When the sheet is folded so that point A rests on the diagonal as shown, the visible shaded area is equal to the visible white area. How many centimeters is A' from its original position, A ? (Source: MATHCOUNTS) **Hints:** 335



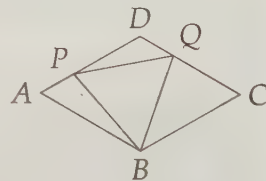
16.35 A rotation about A maps X to Y . A different rotation about X maps A to Y . Show that $\triangle AXY$ is equilateral.

16.36 Point O is inside quadrilateral $ABCD$. A rotation of 180° about O maps the quadrilateral $ABCD$ to itself, meaning each vertex of the image is also a vertex of the original quadrilateral.

- (a) Must $ABCD$ be a parallelogram?
- (b) Must $ABCD$ be a rectangle?
- (c) Must $ABCD$ be a square?

16.37 The image of point A upon a 90 degree clockwise rotation about point B is point C . The image of C upon a 90 degree rotation clockwise about point D is point A . Prove that $ABCD$ is a square.

16.38 Let $ABCD$ be a rhombus with $\angle BAD = 60^\circ$. Let P and Q be points on $\overline{AD}$ and $\overline{CD}$, respectively, such that $\angle PBQ = 60^\circ$. Find the other two angles of triangle PBQ .



16.39 $\triangle ABC$ has $AB = 3$, $BC = 4$, and $AC = 5$.

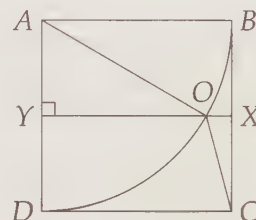
- (a) A cone is formed by rotating the triangle around $\overline{AB}$. What is the volume of this cone?
- (b) A cone is formed by rotating the triangle around $\overline{BC}$. What is the volume of this cone?
- (c)★ A solid is formed by rotating the triangle around $\overline{AC}$. What is the volume of this solid?

16.40 Show that if X' and Y' are the images of X and Y under a dilation with center O and scale factor k , where k is *negative*, then we have $\overline{X'Y'} \parallel \overline{XY}$ and $X'Y'/XY = |k|$.

Challenge Problems

16.41 $\overline{CD}$ is the image of $\overline{AB}$ under some rotation. $\overline{AB}$ and $\overline{CD}$ have the same midpoint, M . Must M be the center of the aforementioned rotation that maps $\overline{AB}$ to $\overline{CD}$? **Hints:** 520

16.42 You might recognize the diagram at right from Problem 8.22 on page 226. It is indeed the same problem. $ABCD$ is a square. The circle with center A and radius AB intersects the perpendicular bisector of $\overline{AD}$ in two points, of which O is the one inside the square. Use the principles we have explored in this chapter to quickly find $\angle AOC$. **Hints:** 241, 180



16.43 Consider the octagon of Problem 16.24.

(a) Show that there is a point inside the octagon that is equidistant from all 8 vertices of the octagon.

Hints: 204

(b) Show that a 90° rotation about the point found in part (a) maps the octagon to itself. **Hints:** 246

16.44 $\triangle ABC$ with vertices $A(2, 4)$, $B(6, 4)$ and $C(4, 10)$ is graphed in a coordinate plane. What will the sum of the abscissas (x -coordinates) of the vertices be when $\triangle ABC$ is reflected over the line $x = 8$? (Source: MATHCOUNTS)

16.45 The image of the incircle of equilateral triangle $\triangle ABC$ upon a dilation about the center of the circle with scale factor k is the circumcircle of $\triangle ABC$. Find k .

16.46 Use a clever dissection of the section of regular octagon $ABCDEFGH$ that is traced in bold at left below to prove that $[ABCDEFGH] = (\sqrt{2})[ACEG]$. (O is the center of the octagon.) **Hints:** 159, 410

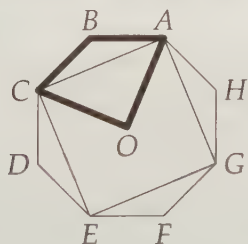


Figure 16.4: Diagram for Problem 16.46

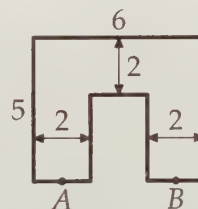


Figure 16.5: Diagram for Problem 16.47

16.47 A laser beam is fired from point A in the U-shaped room at right above. It reflects from each wall as from a perfect mirror. What is the minimum distance the beam travels before hitting a target at point B ? (A and B are the midpoints of their respective sides.) (Source: Mandelbrot)

16.48 Suppose $\overline{AB} \parallel \overline{A'B'}$ and $A'B'/AB = k$, where $k \neq 1$. In this problem, we prove that there is a dilation that maps A to A' and B to B' .

- Why must $\overleftrightarrow{AA'}$ and $\overleftrightarrow{BB'}$ intersect?
- Let P be the intersection of $\overleftrightarrow{AA'}$ and $\overleftrightarrow{BB'}$. Show that $\triangle PAB \sim \triangle PA'B'$.
- Show that there is a dilation with center P that maps A to A' and B to B' .

16.49 Let $ABCD$ be a square, and let M and N be the midpoints of $\overline{AB}$ and $\overline{BC}$, respectively. Prove that $\overline{CM} \perp \overline{DN}$. **Hints:** 445

16.50 The center of the cue ball on my rectangular pool table is directly above point A on the table. I wish to bounce the cue ball off a rail such that after it bounces off the rail, the center of the ball will pass directly over point B on the table. The radius of the cue ball is 1 in. A is 6 inches from the rail, and B is 9 inches from the rail. A is 5 inches from B . How far from the nearest point on the rail to point B do I want the cue ball to hit the rail? **Hints:** 278, 234

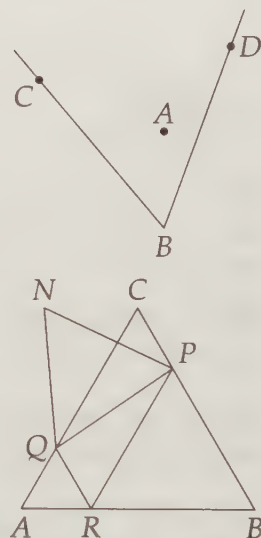
16.51★ What is the total length of the shortest path that goes from $(0, 4)$ to a point on the x -axis, then to a point on the line $y = 6$, then to $(18, 4)$? **Hints:** 262, 472

16.52★ I am standing at point A in the diagram at right. I wish to walk to a point on $\overline{BC}$, then to a point on $\overline{BD}$, then back to my starting point. Construct with straightedge and compass the shortest such path I can take. **Hints:** 338, 370

16.53★ Regular octahedron $ABCDEF$ has volume 360 cubic units. What is the volume of that portion of the octahedron that consists of all points that are closer to A than to any other vertex? **Hints:** 311

16.54★ In the diagram at right, $\triangle ABC$, $\triangle PBR$, $\triangle AQR$, and $\triangle PNQ$ are equilateral. Prove that $BQ = RN = AP$. (Source: Mandelbrot) **Hints:** 320

16.55★ Prove that the image of a circle under rotation about any point is also a circle. **Hints:** 590



Extra! Back on page 430, we introduced the geometric transformation called inversion. Here are some more questions to guide further exploration of inversion:

- What is the image of $\odot O$ upon inversion with respect to $\odot O$?
- What is the image of a line that passes through O ? Of a line that does not pass through O ?
- What is the image of a circle that passes through O ? Of a circle that does not pass through O ?

We'll explore these properties and more in later books in the Art of Problem Solving series.